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Title of chapter: Chapter 08 - Causal Modeling

Summary:

This chapter focuses on how causal modeling moves beyond simple correlations to reveal the underlying cause-and-effect structure in data. It begins by clarifying the distinction between correlation and causation, emphasizing that causal reasoning is central when the goal is to make interventions and predict the outcomes of actions. Directed acyclic graphs (DAGs) are introduced as a key tool to represent causal relations among variables, enabling researchers to visualize dependencies and identify potential confounders. The chapter then walks through the process of causal discovery, where machine learning methods are used to infer causal structures from data, while also highlighting the role of assumptions such as conditional independence.

The second half of the chapter turns to causal inference, quantifying causal effects, particularly in observational settings where experiments are not available. The focus is on estimating the average treatment effect (ATE), with detailed discussion of the assumptions needed for valid inference: no unmeasured confounders (ignorability) and overlap (positivity). Methods for ATE estimation are grouped into treatment models (like propensity score methods) and outcome models (like regression-based approaches), both of which rely on machine learning to adjust for confounding. The chapter emphasizes the iterative process of specifying a causal method, selecting an appropriate machine learning model, training it, and checking whether assumptions hold. Together, these steps equip practitioners with the tools to answer "what if" questions responsibly in applied domains such as healthcare or social policy.

What are two points that you learned after reading this chapter?:

- Causal modeling is not just about discovering relationships but also requires strong assumptions (ignorability and overlap) for valid causal effect estimation from observational data.
- Average treatment effect estimation can be approached through two main strategies: treatment models (e.g., propensity score matching, weighting) and outcome models (e.g., regression, machine learning predictors), both of which have different strengths and weaknesses depending on the data structure.

What are two questions that you have about this chapter?:

- How can researchers be confident that they have included all relevant confounders in their dataset when the "no unmeasured confounders" assumption is so critical but often unverifiable?
- When overlap is limited (e.g., certain groups are almost always treated or untreated), what methods can be used to still make causal inferences without discarding too much data?

Do you have some initial thoughts or insights to approach the questions?:

- To address unmeasured confounding, researchers can use domain expertise, sensitivity analysis, and instrumental variable approaches to assess the robustness of their conclusions. While full certainty is impossible, triangulating results across different models helps.
- For limited overlap, methods like trimming (restricting to regions of overlap), weighting adjustments, or doubly robust estimators may help retain interpretability while reducing bias. In some cases, partial causal effects (local ATEs) may be more defensible than trying to generalize across the whole population.